Visual Document Retrieval Pipeline

Two Approaches for PDF Question Answering

Overview

Problem Statement

- PDF documents containing images and complex layouts
- Need to answer user questions accurately
- Challenge: Preserve visual information vs. text extraction

Two Main Approaches

- **Approach A:** Visual Document Retrieval (Image-First)

  –

  Optional: DSPy Query Enhancement
- **Approach B:** VLLM Description + Text Filtering (Hybrid)

Approach A - Visual Document Retrieval

End-to-End Visual Pipeline

1 Visual Retrieval

- User submits a question
- *(Optional: DSPy query refinement)*
- Retrieve top-k relevant pages using vision models:
  - **ColPali** - Document understanding model
  - **ColQwen** - Multimodal retrieval model
- Pages ranked by visual + textual relevance

2 VLLM Processing

- Top-k pages sent directly as images
- Processed by state-of-the-art VLLMs

3 Answer Generation

- Image versions of pages used as input
- *(Optional: Enhanced generation query)*
- Preserves layouts, charts, tables, diagrams

Approach A - Supported Models

Vision-Language Models (VLLMs)

Current Generation

- ✓ GPT-4o
- ✓ GPT-4.1 / GPT-4.1-mini / GPT-4.1-nano

Advanced Reasoning

- ✓ o3
- ✓ o4-mini

Next Generation

- ⭐ **GPT-5**

  (Recommended for DSPy-enhanced pipeline)
- ✓ GPT-5 mini

Note: *All models support native image understanding*

Approach A - DSPy Enhancement (Optional)

Query Refinement for Better Retrieval & Generation

What is DSPy?

- Declarative Self-improving Python framework
- Automatic prompt optimization
- Improves both retrieval and generation quality

Enhancement Pipeline

```
Original Query → DSPy Refinement → Enhanced Retrieval Query
→ Enhanced Generation Query
```

Key Benefits

🎯 More precise retrieval results

📝 Better-structured generation prompts

🔄 Systematic query optimization

⚡ Improved end-to-end accuracy

DSPy-Enhanced Pipeline Flow

Approach A with Query Refinement

0 Query Enhancement (DSPy)

User Question:

`"What were the revenue trends in Q3?"`

↓ DSPy Processing ↓

Enhanced Retrieval Query:

`"Revenue trends, Q3 financial data, quarterly comparisons, revenue charts, sales figures third quarter"`

Enhanced Generation Query:

`"Analyze the Q3 revenue trends shown in the retrieved pages. Include: year-over-year comparison, key metrics, visual chart interpretation, and notable changes."`

1 Visual Retrieval

- **ColPali** uses enhanced retrieval query
- Selects top-k pages with improved precision
- Better semantic matching

2 Generation

- Top-k pages sent to **GPT-5**
- Enhanced generation query provides better context
- More focused and accurate answers

Approach A - Key Advantages

Why Visual-First?

✓ Preserves Visual Context

- Charts, graphs, and diagrams analyzed directly
- Layout and formatting maintained
- Handwritten notes and annotations captured

✓ Simplified Pipeline

- Fewer transformation steps
- No OCR errors
- Direct visual understanding

✓ Better for Complex Documents

- Technical drawings
- Infographics
- Multi-column layouts

Approach B - VLLM Description Pipeline

Hybrid Text-Visual Approach

1

Page Description

- Each PDF page converted to image
- VLLM generates text description of every page
- Creates searchable text corpus

2

Text Filtering (Two Methods)

2a) Keyword Filtering

– Filter pages by domain-specific keywords
– Boolean matching
– Fast preprocessing

2b) Embedding Search

– Semantic similarity search
– Vector representations
– Context-aware matching

Approach B - Answer Generation

Two Output Strategies

Option 3a: Text-Based Input

Question → Text Filtering → Relevant Page Descriptions → Text LLM → Answer

- Uses text descriptions from Step 1
- Faster processing
- Lower computational cost
- Suitable for text-heavy documents

Option 3b: Image-Based Input

Question → Text Filtering → Relevant Page Images → VLLM → Answer

- Retrieves original page images
- Preserves visual information
- Best of both worlds

Approach Comparison

Visual vs. Hybrid Pipeline

Aspect	Approach A (Visual)	Approach B (Hybrid)
Retrieval	ColPali/ColQwen	Text embeddings/keywords
Initial Processing	None	VLLM page description
Filtering Speed	Model-dependent	Fast (text-based)
Visual Preservation	Complete	Optional (3b only)
Setup Complexity	Low	Medium (requires preprocessing)
Best For	Visual-heavy docs	Mixed content

Use Case Recommendations

When to Use Each Approach

Approach A (Visual Document Retrieval)

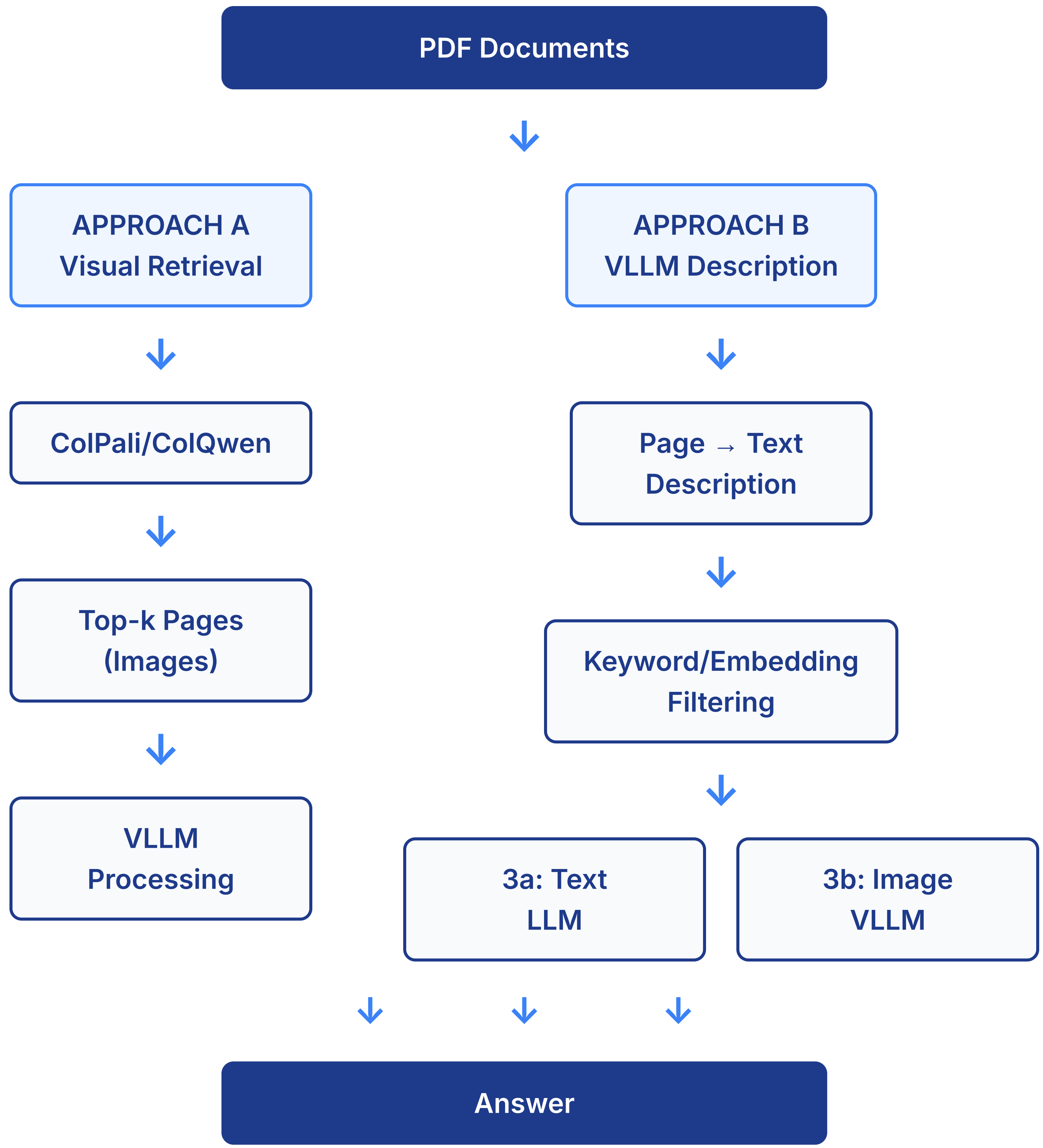
- 📊 Financial reports with charts
- 📐 Technical drawings and blueprints
- 📈 Data visualization dashboards
- 📝 Forms with mixed content
- 🎨 Marketing materials

Approach B (VLLM Description)

- 📄 Long text-heavy documents
- ⚡ Need for fast keyword filtering
- 🔍 Pre-indexed document libraries
- 💰 Cost-sensitive applications (3a variant)
- 🔄 Frequently searched document sets

Pipeline Architecture

Visual Representation



Performance Considerations

Optimization Factors

Latency

- **Approach A:** Single retrieval + VLLM call
- **Approach B:** Preprocessing time + filtering + LLM call

Accuracy

- **Approach A:** Excellent for visual content
- **Approach B-3a:** Best for text extraction
- **Approach B-3b:** Balanced approach

Cost

- **Approach A:** VLLM costs per query
- **Approach B:** Upfront preprocessing + per-query costs

Scalability

- **Approach A:** Scales with query volume
- **Approach B:** Better for large, static corpora

Implementation Recommendations

For New Projects

1. Start with **Approach A** for simplicity
2. Evaluate on representative documents
3. Add **Approach B** if needed for scale

For Production Systems

1. Use **Approach B-3b** as baseline
2. Implement both filtering methods (keywords + embeddings)
3. A/B test visual vs. hybrid retrieval

Hybrid Strategy

- Route queries based on document type
  - Visual docs → Approach A
  - Text docs → Approach B-3a
  - Mixed docs → Approach B-3b

Future Enhancements

Potential Improvements

🔄 Ensemble Methods

- Combine ColPali/ColQwen with text embeddings
- Re-ranking strategies

🎯 Adaptive Routing

- ML model to select best approach per query
- Document type classification

📊 Feedback Loop

- Track answer quality
- Refine retrieval parameters
- Model fine-tuning

🚀 Advanced Models

- Leverage GPT-5 capabilities
- Specialized domain models

Conclusion

Two Powerful Approaches

- **Visual-First (A):** Best for preserving document context
- **Hybrid (B):** Flexible with filtering options

Key Decision Factors

- Document characteristics
- Performance requirements
- Cost constraints
- Accuracy needs

Recommendation

- Prototype both approaches
- Measure on your specific use case
- Consider hybrid deployment

Thank You!

Key Takeaways

1. Visual document retrieval preserves context
2. Hybrid approaches offer flexibility
3. Choose based on document types and requirements
4. Both pipelines support state-of-the-art VLLMs

Questions?

Appendix: Technical Details

ColPali

- Vision-language retrieval model
- Optimized for document understanding
- Handles multiple languages

ColQwen

- Multimodal retrieval
- Strong visual grounding
- Efficient embedding generation

VLLM Models

- GPT-4o: Optimized multimodal
- o3/o4-mini: Advanced reasoning
- GPT-5: Next-generation capabilities